

Hilbert Spaces

Functional Analysis

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Definition

*Two elements of an inner product space are said to be **orthogonal** if their inner product equals 0.*

Theorem (Pythagorean Theorem)

If u, v are orthogonal elements of an inner product space, then

$$\|f + g\|^2 = \|f\|^2 + \|g\|^2.$$

Theorem (Cauchy-Schwarz Inequality)

Suppose f and g are elements of an inner product space. Then

$$|\langle f, g \rangle| \leq \|f\| \|g\|$$

Theorem (Triangle Inequality)

Suppose f and g are elements of an inner product space. Then

$$\|f + g\| \leq \|f\| + \|g\|$$

with equality if and only if one of f, g is a nonnegative multiple of the other.

Theorem (Inner products induce norms)

If V is an inner product space, then

$$\|\cdot\| = \langle \cdot, \cdot \rangle$$

defines a norm on V .

Theorem (Parallelogram Equality)

Suppose f and g are elements of an inner product space. Then

$$\|f + g\|^2 + \|f - g\|^2 = 2\|f\|^2 + 2\|g\|^2.$$

Definition

A **Hilbert space** is an inner product space that is a Banach space with the norm induced by the inner product.

Definition

Suppose U is a nonempty subset of a normed space V and let $f \in V$. The distance from f to U , denoted $\text{distance}(f, U)$, is defined by

$$\text{distance}(f, U) = \inf\{\|f - g\| : g \in U\}.$$

Definition

A subset of a linear space is said to be **convex** if

$$(1 - t)f + tg \in U \text{ for all } t \in [0, 1] \text{ and all } f, g \in U.$$

This means that U contains the line segment connecting any pair of points in U .

Theorem

If V is a Hilbert space, $f \in V$, and U is a nonempty closed convex subset of V . Then there exists a unique $g \in U$ such that

$$\|f - g\| = \text{distance}(f, U).$$

Definition

*Suppose U is a nonempty closed convex subset of a Hilbert space V . The **orthogonal projection** of V onto U is the function $P_U \in \text{End}(V)$ defined by setting $P_U(f)$ equal to the (unique) element of U closest to f .*

Theorem

Suppose U is a closed subspace of a Hilbert space V and $f \in V$. Then

- i) $f - P_U f$ is orthogonal to g for every $g \in U$;*
- ii) if $h \in U$ and $f - h$ is orthogonal to each $g \in U$, then $h = P_U f$;*
- iii) $P_U : V \rightarrow V$ is a linear map;*
- iv) $\|P_U f\| \leq \|f\|$, with equality if and only if $f \in U$.*

Definition

Suppose U is a subset of an inner product space. The **orthogonal complement** of U , denoted by $U^\perp$, is defined as

$$U^\perp = \{h \in V : \langle h, g \rangle = 0 \text{ for all } g \in U\}.$$

Theorem (Properties of the Orthogonal Complement)

Suppose U is an inner product space V . Then

- i) $U^\perp$ is a closed subspace of V ;
- ii) $U \cap U^\perp \subseteq \{0\}$;
- iii) if $W \subseteq U$, then $U^\perp \subseteq W^\perp$;
- iv) $\overline{U}^\perp = U^\perp$;
- v) $U \subseteq (U^\perp)^\perp$.

Theorem

Suppose U is a subspace of a Hilbert space V . Then

$$\overline{U} = (U^\perp)^\perp.$$

Theorem

Suppose U is a subspace of a Hilbert space V . Then

$$\overline{U} = V \text{ if and only if } U^\perp = \{0\}.$$

Theorem

Suppose U is a closed subspace of a Hilbert space V . Then every element $f \in V$ can be uniquely written in the form

$$f = g + h,$$

where $g \in U$ and $h \in U^\perp$. Furthermore, $g = P_U f$ and $h = f - P_U f$.

Theorem

Suppose U is a closed subspace of a Hilbert space V . Then

- i) $\operatorname{im} P_U = U$ and $\ker P_U = U^\perp$;
- ii) $\operatorname{im} P_{U^\perp} = U^\perp$ and $\ker P_{U^\perp} = U$;
- iii) $P_U = \operatorname{Id} - P_{U^\perp}$.

Theorem (Riesz Representation Theorem)

Suppose φ is a linear functional on a Hilbert space V . Then there exists a unique $h \in V$ such that

$$\varphi(f) = \langle f, h \rangle$$

for all $f \in V$. Furthermore, $\|\varphi\| = \|h\|$.

Definition

A family $\{e_k\}_{k \in \Gamma}$ in an inner product space is called an **orthonormal family** if

$$\langle e_i, e_j \rangle = \delta_{ij}$$

for all $i, j \in \Gamma$.

Example

Some examples of orthonormal families:

- Let e_k be the element of ℓ^2 all of whose coordinates are 0 except for the k^{th} coordinate, which is 1. Then $\{e_k\}_{k \in \mathbb{N}}$ is an orthonormal family in ℓ^2 .
- For $k \in \mathbb{Z}$, define $e_k : (-\pi, \pi]$ by

$$e_k(t) = \begin{cases} \frac{1}{\sqrt{\pi}} \sin kt & k > 0, \\ 1 & k = 0, \\ \frac{1}{\sqrt{2\pi}} & \\ \frac{1}{\sqrt{\pi}} \cos kt & k < 0. \end{cases}$$

Then $\{e_k\}_{k \in \mathbb{Z}}$ is an orthonormal family in $L^2((-\pi, \pi])$.

Theorem

Suppose Ω is a finite set and $\{e_j\}_{j \in \Omega}$ is an orthonormal family in an inner product space. Then

$$\left\| \sum_{j \in \Omega} \alpha_j e_j \right\|^2 = \sum_{j \in \Omega} |\alpha_j|^2$$

for every family $\{\alpha_j\}_{j \in \Omega}$ in $\mathbb{C}$.

Remark

*Linear algebra, and algebra more generally, deals with sums of only finitely many terms. However, in analysis we often want to sum infinitely many terms. For example, earlier we defined the infinite sum of a sequence (g_n) in a normed vector space to be the limit of the partial sums $s_n = \sum_{k=1}^n g_k$ if that limit exists. The next definition captures a more powerful method for dealing with infinite sums. The sum defined below is called an **unordered sum** because the set Γ is not assumed to come with any ordering.*

Definition

Suppose $\{f_k\}_{k \in \Gamma}$ is a family in a normed linear space V . The **unordered sum** $\sum_{k \in \Gamma} f_k$ is said to **converge** if there exists $g \in V$ such that for every $\varepsilon > 0$, there exists a finite subset Ω of Γ such that

$$\left\| \sum_{j \in \Omega'} f_j - g \right\| < \varepsilon$$

for all finite sets Ω' with $\Omega \subseteq \Omega' \subseteq \Gamma$. If this happens, we set $\sum_{k \in \Gamma} f_k = g$, otherwise $\sum_{k \in \Gamma} f_k$ is left undefined.

Theorem

Suppose $\{e_k\}_{k \in \Gamma}$ is an orthonormal family in a Hilbert space V . Suppose $\{\alpha_k\}_{k \in \Gamma}$ is a family in $\mathbb{C}$. Then

- i) The unordered sum $\sum_{k \in \Gamma} \alpha_k e_k$ converges if and only if $\sum_{k \in \Gamma} |\alpha_k|^2 < \infty$.

Furthermore, if $\sum_{k \in \Gamma} \alpha_k e_k$ converges, then

ii)
$$\left\| \sum_{k \in \Gamma} \alpha_k e_k \right\|^2 = \sum_{k \in \Gamma} |\alpha_k|^2.$$

Theorem (Bessel's Inequality)

Suppose $\{e_k\}_{k \in \Gamma}$ is an orthonormal family in an inner product space V and $f \in V$. Then

$$\sum_{k \in \Gamma} |\langle f, e_k \rangle|^2 \leq \|f\|^2$$

Theorem

Suppose $\{e_k\}_{k \in \Gamma}$ is an orthonormal family in a Hilbert space V . Then

$$\text{i) } \overline{\text{span}\{e_k\}_{k \in \Gamma}} = \left\{ \sum_{k \in \Gamma} \alpha_k e_k : \{\alpha_k\}_{k \in \Gamma} \text{ is a family in } \mathbb{C} \text{ and } \sum_{k \in \Gamma} |\alpha_k|^2 < \infty \right\}.$$

Furthermore,

$$\text{ii) } f = \sum_{k \in \Gamma} \langle f, e_k \rangle e_k$$

for every $f \in \overline{\text{span}\{e_k\}_{k \in \Gamma}}$.

Definition

An orthonormal family $\{e_k\}_{k \in \Gamma}$ in a Hilbert space V is called an **orthonormal basis** of V if

$$\overline{\text{span}\{e_k\}_{k \in \Gamma}} = V.$$

Theorem (Parseval's Identity)

Suppose $\{e_k\}_{k \in \Gamma}$ is an orthonormal basis of a Hilbert space V and $f, g \in V$. Then

- i) $f = \sum_{k \in \Gamma} \langle f, e_k \rangle e_k;$
- ii) $\langle f, g \rangle = \sum_{k \in \Gamma} \langle f, e_k \rangle \langle e_k, g \rangle;$
- iii) $\|f\|^2 = \sum_{k \in \Gamma} |\langle f, e_k \rangle|^2.$

Definition

A normed linear space is called **separable** if it has a countable dense subset.

Theorem

Every separable Hilbert space has an orthonormal basis. More generally, every Hilbert space has a basis.

Theorem (Generalised Riesz Representation Theorem)

Suppose φ is a bounded linear functional on a Hilbert space V and $\{e_k\}_{k \in \Gamma}$ is an orthonormal basis of V . Let

$$h = \sum_{k \in \Gamma} \varphi(e_k)^* e_k.$$

Then

$$\varphi(f) = \langle f, h \rangle$$

for all $f \in V$. Furthermore, $\|\varphi\| = (\sum_{k \in \Gamma} |\varphi(e_k)|^2)^{1/2}$.